40

File System Implementation

In this chapter, we introduce a simple file system implementation, known as **vsfs** (the **Very Simple File System**). This file system is a simplified version of a typical UNIX file system and thus serves to introduce some of the basic on-disk structures, access methods, and various policies that you will find in many file systems today.

The file system is pure software; unlike our development of CPU and memory virtualization, we will not be adding hardware features to make some aspect of the file system work better (though we will want to pay attention to device characteristics to make sure the file system works well). Because of the great flexibility we have in building a file system, many different ones have been built, literally from AFS (the Andrew File System) [H+88] to ZFS (Sun's Zettabyte File System) [B07]. All of these file systems have different data structures and do some things better or worse than their peers. Thus, the way we will be learning about file systems is through case studies: first, a simple file system (vsfs) in this chapter to introduce most concepts, and then a series of studies of real file systems to understand how they can differ in practice.

THE CRUX: HOW TO IMPLEMENT A SIMPLE FILE SYSTEM

How can we build a simple file system? What structures are needed on the disk? What do they need to track? How are they accessed?

40.1 The Way To Think

To think about file systems, we usually suggest thinking about two different aspects of them; if you understand both of these aspects, you probably understand how the file system basically works.

The first is the **data structures** of the file system. In other words, what types of on-disk structures are utilized by the file system to organize its data and metadata? The first file systems we'll see (including vsfs below) employ simple structures, like arrays of blocks or other objects, whereas

ASIDE: MENTAL MODELS OF FILE SYSTEMS

As we've discussed before, mental models are what you are really trying to develop when learning about systems. For file systems, your mental model should eventually include answers to questions like: what on-disk structures store the file system's data and metadata? What happens when a process opens a file? Which on-disk structures are accessed during a read or write? By working on and improving your mental model, you develop an abstract understanding of what is going on, instead of just trying to understand the specifics of some file-system code (though that is also useful, of course!).

more sophisticated file systems, like SGI's XFS, use more complicated tree-based structures [S+96].

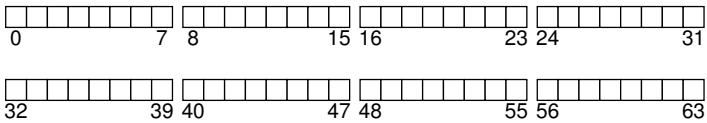
The second aspect of a file system is its **access methods**. How does it map the calls made by a process, such as `open()`, `read()`, `write()`, etc., onto its structures? Which structures are read during the execution of a particular system call? Which are written? How efficiently are all of these steps performed?

If you understand the data structures and access methods of a file system, you have developed a good mental model of how it truly works, a key part of the systems mindset. Try to work on developing your mental model as we delve into our first implementation.

40.2 Overall Organization

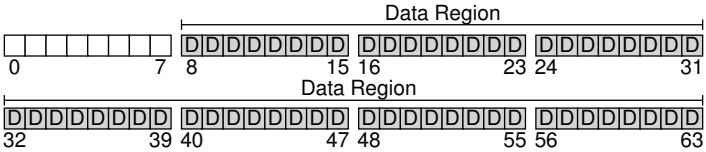
We now develop the overall on-disk organization of the data structures of the vsfs file system. The first thing we'll need to do is divide the disk into **blocks**; simple file systems use just one block size, and that's exactly what we'll do here. Let's choose a commonly-used size of 4 KB.

Thus, our view of the disk partition where we're building our file system is simple: a series of blocks, each of size 4 KB. The blocks are addressed from 0 to $N-1$, in a partition of size N 4-KB blocks. Assume we have a really small disk, with just 64 blocks:



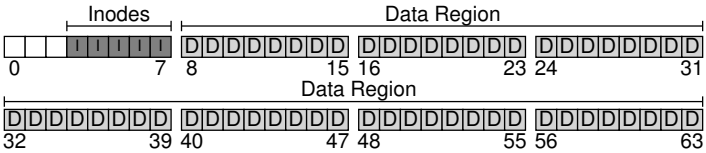
Let's now think about what we need to store in these blocks to build a file system. Of course, the first thing that comes to mind is user data. In fact, most of the space in any file system is (and should be) user data. Let's call the region of the disk we use for user data the **data region**, and,

again for simplicity, reserve a fixed portion of the disk for these blocks, say the last 56 of 64 blocks on the disk:



As we learned about (a little) last chapter, the file system has to track information about each file. This information is a key piece of **metadata**, and tracks things like which data blocks (in the data region) comprise a file, the size of the file, its owner and access rights, access and modify times, and other similar kinds of information. To store this information, file systems usually have a structure called an **inode** (we'll read more about inodes below).

To accommodate inodes, we'll need to reserve some space on the disk for them as well. Let's call this portion of the disk the **inode table**, which simply holds an array of on-disk inodes. Thus, our on-disk image now looks like this picture, assuming that we use 5 of our 64 blocks for inodes (denoted by I's in the diagram):

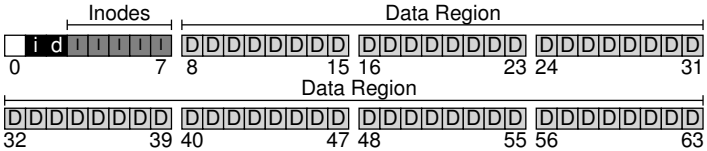


We should note here that inodes are typically not that big, for example 128 or 256 bytes. Assuming 256 bytes per inode, a 4-KB block can hold 16 inodes, and our file system above contains 80 total inodes. In our simple file system, built on a tiny 64-block partition, this number represents the maximum number of files we can have in our file system; however, do note that the same file system, built on a larger disk, could simply allocate a larger inode table and thus accommodate more files.

Our file system thus far has data blocks (D), and inodes (I), but a few things are still missing. One primary component that is still needed, as you might have guessed, is some way to track whether inodes or data blocks are free or allocated. Such **allocation structures** are thus a requisite element in any file system.

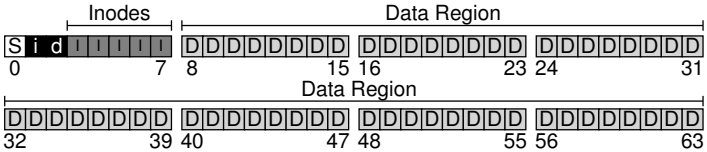
Many allocation-tracking methods are possible, of course. For example, we could use a **free list** that points to the first free block, which then points to the next free block, and so forth. We instead choose a simple and popular structure known as a **bitmap**, one for the data region (the **data bitmap**), and one for the inode table (the **inode bitmap**). A bitmap is a

simple structure: each bit is used to indicate whether the corresponding object/block is free (0) or in-use (1). And thus our new on-disk layout, with an inode bitmap (i) and a data bitmap (d):



You may notice that it is a bit of overkill to use an entire 4-KB block for these bitmaps; such a bitmap can track whether 32K objects are allocated, and yet we only have 80 inodes and 56 data blocks. However, we just use an entire 4-KB block for each of these bitmaps for simplicity.

The careful reader (i.e., the reader who is still awake) may have noticed there is one block left in the design of the on-disk structure of our very simple file system. We reserve this for the **superblock**, denoted by an S in the diagram below. The superblock contains information about this particular file system, including, for example, how many inodes and data blocks are in the file system (80 and 56, respectively in this instance), where the inode table begins (block 3), and so forth. It will likely also include a magic number of some kind to identify the file system type (in this case, vsfs).



Thus, when mounting a file system, the operating system will read the superblock first, to initialize various parameters, and then attach the volume to the file-system tree. When files within the volume are accessed, the system will thus know exactly where to look for the needed on-disk structures.

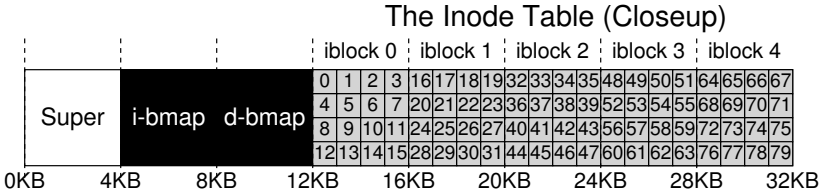
40.3 File Organization: The Inode

One of the most important on-disk structures of a file system is the **inode**; virtually all file systems have a structure similar to this. The name inode is short for **index node**, the historical name given to it in UNIX [RT74] and possibly earlier systems, used because these nodes were originally arranged in an array, and the array *indexed* into when accessing a particular inode.

ASIDE: DATA STRUCTURE — THE INODE

The **inode** is the generic name that is used in many file systems to describe the structure that holds the metadata for a given file, such as its length, permissions, and the location of its constituent blocks. The name goes back at least as far as UNIX (and probably further back to Multics if not earlier systems); it is short for **index node**, as the inode number is used to index into an array of on-disk inodes in order to find the inode of that number. As we'll see, design of the inode is one key part of file system design. Most modern systems have some kind of structure like this for every file they track, but perhaps call them different things (such as dnodes, fnodes, etc.).

Each inode is implicitly referred to by a number (called the **i-number**), which we've earlier called the **low-level name** of the file. In vsfs (and other simple file systems), given an i-number, you should directly be able to calculate where on the disk the corresponding inode is located. For example, take the inode table of vsfs as above: 20KB in size (five 4KB blocks) and thus consisting of 80 inodes (assuming each inode is 256 bytes); further assume that the inode region starts at 12KB (i.e, the superblock starts at 0KB, the inode bitmap is at address 4KB, the data bitmap at 8KB, and thus the inode table comes right after). In vsfs, we thus have the following layout for the beginning of the file system partition (in closeup view):



To read inode number 32, the file system would first calculate the offset into the inode region ($32 \cdot sizeof(inode)$ or 8192), add it to the start address of the inode table on disk (`inodeStartAddr` = $12KB$), and thus arrive upon the correct byte address of the desired block of inodes: $20KB$. Recall that disks are not byte addressable, but rather consist of a large number of addressable sectors, usually 512 bytes. Thus, to fetch the block of inodes that contains inode 32, the file system would issue a read to sector $\frac{20 \times 1024}{512}$, or 40, to fetch the desired inode block. More generally, the sector address `sector` of the inode block can be calculated as follows:

```
blk    = (inumber * sizeof(inode_t)) / blockSize;
sector = ((blk * blockSize) + inodeStartAddr) / sectorSize;
```

Inside each inode is virtually all of the information you need about a file: its *type* (e.g., regular file, directory, etc.), its *size*, the number of *blocks* allocated to it, *protection information* (such as who owns the file, as well

Size	Name	What is this inode field for?
2	mode	can this file be read/written/executed?
2	uid	who owns this file?
4	size	how many bytes are in this file?
4	time	what time was this file last accessed?
4	ctime	what time was this file created?
4	mtime	what time was this file last modified?
4	dtime	what time was this inode deleted?
2	gid	which group does this file belong to?
2	links_count	how many hard links are there to this file?
4	blocks	how many blocks have been allocated to this file?
4	flags	how should ext2 use this inode?
4	osd1	an OS-dependent field
60	block	a set of disk pointers (15 total)
4	generation	file version (used by NFS)
4	file_acl	a new permissions model beyond mode bits
4	dir_acl	called access control lists

Figure 40.1: **Simplified Ext2 Inode**

as who can access it), some *time* information, including when the file was created, modified, or last accessed, as well as information about where its data blocks reside on disk (e.g., pointers of some kind). We refer to all such information about a file as **metadata**; in fact, any information inside the file system that isn't pure user data is often referred to as such. An example inode from ext2 [P09] is shown in Figure 40.1[1].

One of the most important decisions in the design of the inode is how it refers to where data blocks are. One simple approach would be to have one or more **direct pointers** (disk addresses) inside the inode; each pointer refers to one disk block that belongs to the file. Such an approach is limited: for example, if you want to have a file that is really big (e.g., bigger than the block size multiplied by the number of direct pointers in the inode), you are out of luck.

The Multi-Level Index

To support bigger files, file system designers have had to introduce different structures within inodes. One common idea is to have a special pointer known as an **indirect pointer**. Instead of pointing to a block that contains user data, it points to a block that contains more pointers, each of which point to user data. Thus, an inode may have some fixed number of direct pointers (e.g., 12), and a single indirect pointer. If a file grows large enough, an indirect block is allocated (from the data-block region of the disk), and the inode's slot for an indirect pointer is set to point to it. Assuming 4-KB blocks and 4-byte disk addresses, that adds another 1024 pointers; the file can grow to be $(12 + 1024) \cdot 4K$ or 4144KB.

[1] Type info is kept in the directory entry, and thus is not found in the inode itself.

TIP: CONSIDER EXTENT-BASED APPROACHES

A different approach is to use **extents** instead of pointers. An extent is simply a disk pointer plus a length (in blocks); thus, instead of requiring a pointer for every block of a file, all one needs is a pointer and a length to specify the on-disk location of a file. Just a single extent is limiting, as one may have trouble finding a contiguous chunk of on-disk free space when allocating a file. Thus, extent-based file systems often allow for more than one extent, thus giving more freedom to the file system during file allocation.

In comparing the two approaches, pointer-based approaches are the most flexible but use a large amount of metadata per file (particularly for large files). Extent-based approaches are less flexible but more compact; in particular, they work well when there is enough free space on the disk and files can be laid out contiguously (which is the goal for virtually any file allocation policy anyhow).

Not surprisingly, in such an approach, you might want to support even larger files. To do so, just add another pointer to the inode: the **double indirect pointer**. This pointer refers to a block that contains pointers to indirect blocks, each of which contain pointers to data blocks. A double indirect block thus adds the possibility to grow files with an additional $1024 \cdot 1024$ or 1-million 4KB blocks, in other words supporting files that are over 4GB in size. You may want even more, though, and we bet you know where this is headed: the **triple indirect pointer**.

Overall, this imbalanced tree is referred to as the **multi-level index** approach to pointing to file blocks. Let's examine an example with twelve direct pointers, as well as both a single and a double indirect block. Assuming a block size of 4 KB, and 4-byte pointers, this structure can accommodate a file of just over 4 GB in size (i.e., $(12 + 1024 + 1024^2) \times 4\ KB$). Can you figure out how big of a file can be handled with the addition of a triple-indirect block? (hint: pretty big)

Many file systems use a multi-level index, including commonly-used file systems such as Linux ext2 [P09] and ext3, NetApp's WAFL, as well as the original UNIX file system. Other file systems, including SGI XFS and Linux ext4, use **extents** instead of simple pointers; see the earlier aside for details on how extent-based schemes work (they are akin to segments in the discussion of virtual memory).

You might be wondering: why use an imbalanced tree like this? Why not a different approach? Well, as it turns out, many researchers have studied file systems and how they are used, and virtually every time they find certain "truths" that hold across the decades. One such finding is that *most files are small*. This imbalanced design reflects such a reality; if most files are indeed small, it makes sense to optimize for this case. Thus, with a small number of direct pointers (12 is a typical number), an inode

Most files are small	~2K is the most common size
Average file size is growing	Almost 200K is the average
Most bytes are stored in large files	A few big files use most of space
File systems contain lots of files	Almost 100K on average
File systems are roughly half full	Even as disks grow, file systems remain ~50% full
Directories are typically small	Many have few entries; most have 20 or fewer

Figure 40.2: **File System Measurement Summary**

can directly point to 48 KB of data, needing one (or more) indirect blocks for larger files. See Agrawal et. al [A+07] for a recent-ish study; Figure 40.2 summarizes those results.

Of course, in the space of inode design, many other possibilities exist; after all, the inode is just a data structure, and any data structure that stores the relevant information, and can query it effectively, is sufficient. As file system software is readily changed, you should be willing to explore different designs should workloads or technologies change.

40.4 Directory Organization

In vsfs (as in many file systems), directories have a simple organization; a directory basically just contains a list of (entry name, inode number) pairs. For each file or directory in a given directory, there is a string and a number in the data block(s) of the directory. For each string, there may also be a length (assuming variable-sized names).

For example, assume a directory `dir` (inode number 5) has three files in it (`foo`, `bar`, and `foobar_is_a_pretty_longname`), with inode numbers 12, 13, and 24 respectively. The on-disk data for `dir` might look like:

```
inum | reclen | strlen | name
  5      12        2     .
  2      12        3     ..
 12      12        4     foo
 13      12        4     bar
 24      36       28     foobar_is_a_pretty_longname
```

In this example, each entry has an inode number, record length (the total bytes for the name plus any left over space), string length (the actual length of the name), and finally the name of the entry. Note that each directory has two extra entries, `.` "dot" and `..` "dot-dot"; the dot directory is just the current directory (in this example, `dir`), whereas dot-dot is the parent directory (in this case, the root).

Deleting a file (e.g., calling `unlink()`) can leave an empty space in the middle of the directory, and hence there should be some way to mark that as well (e.g., with a reserved inode number such as zero). Such a delete is one reason the record length is used: a new entry may reuse an old, bigger entry and thus have extra space within.

ASIDE: LINKED-BASED APPROACHES

Another simpler approach in designing inodes is to use a **linked list**. Thus, inside an inode, instead of having multiple pointers, you just need one, to point to the first block of the file. To handle larger files, add another pointer at the end of that data block, and so on, and thus you can support large files.

As you might have guessed, linked file allocation performs poorly for some workloads; think about reading the last block of a file, for example, or just doing random access. Thus, to make linked allocation work better, some systems will keep an in-memory table of link information, instead of storing the next pointers with the data blocks themselves. The table is indexed by the address of a data block D; the content of an entry is simply D's next pointer, i.e., the address of the next block in a file which follows D. A null-value could be there too (indicating an end-of-file), or some other marker to indicate that a particular block is free. Having such a table of next pointers makes it so that a linked allocation scheme can effectively do random file accesses, simply by first scanning through the (in memory) table to find the desired block, and then accessing (on disk) it directly.

Does such a table sound familiar? What we have described is the basic structure of what is known as the **file allocation table**, or **FAT** file system. Yes, this classic old Windows file system, before NTFS [C94], is based on a simple linked-based allocation scheme. There are other differences from a standard UNIX file system too; for example, there are no inodes per se, but rather directory entries which store metadata about a file and refer directly to the first block of said file, which makes creating hard links impossible. See Brouwer [B02] for more of the inelegant details.

You might be wondering where exactly directories are stored. Often, file systems treat directories as a special type of file. Thus, a directory has an inode, somewhere in the inode table (with the type field of the inode marked as "directory" instead of "regular file"). The directory has data blocks pointed to by the inode (and perhaps, indirect blocks); these data blocks live in the data block region of our simple file system. Our on-disk structure thus remains unchanged.

We should also note again that this simple linear list of directory entries is not the only way to store such information. As before, any data structure is possible. For example, XFS [S+96] stores directories in B-tree form, making file create operations (which have to ensure that a file name has not been used before creating it) faster than systems with simple lists that must be scanned in their entirety.

ASIDE: FREE SPACE MANAGEMENT

There are many ways to manage free space; bitmaps are just one way. Some early file systems used **free lists**, where a single pointer in the super block was kept to point to the first free block; inside that block the next free pointer was kept, thus forming a list through the free blocks of the system. When a block was needed, the head block was used and the list updated accordingly.

Modern file systems use more sophisticated data structures. For example, SGI's XFS [S+96] uses some form of a **B-tree** to compactly represent which chunks of the disk are free. As with any data structure, different time-space trade-offs are possible.

40.5 Free Space Management

A file system must track which inodes and data blocks are free, and which are not, so that when a new file or directory is allocated, it can find space for it. Thus **free space management** is important for all file systems. In vsfs, we have two simple bitmaps for this task.

For example, when we create a file, we will have to allocate an inode for that file. The file system will thus search through the bitmap for an inode that is free, and allocate it to the file; the file system will have to mark the inode as used (with a 1) and eventually update the on-disk bitmap with the correct information. A similar set of activities take place when a data block is allocated.

Some other considerations might also come into play when allocating data blocks for a new file. For example, some Linux file systems, such as ext2 and ext3, will look for a sequence of blocks (say 8) that are free when a new file is created and needs data blocks; by finding such a sequence of free blocks, and then allocating them to the newly-created file, the file system guarantees that a portion of the file will be contiguous on the disk, thus improving performance. Such a **pre-allocation** policy is thus a commonly-used heuristic when allocating space for data blocks.

40.6 Access Paths: Reading and Writing

Now that we have some idea of how files and directories are stored on disk, we should be able to follow the flow of operation during the activity of reading or writing a file. Understanding what happens on this **access path** is thus the second key in developing an understanding of how a file system works; pay attention!

For the following examples, let us assume that the file system has been mounted and thus that the superblock is already in memory. Everything else (i.e., inodes, directories) is still on the disk.

	data bitmap	inode bitmap	root inode	foo inode	bar inode	root data	foo data	bar data [0]	bar data [1]	bar data [2]
open(bar)			**read**							
						read				
				read						
							read			
					read					
read()					**read**					
								read		
					write					
read()					**read**					
									read	
					write					
read()					**read**					
										read
					write					

Figure 40.3: **File Read Timeline (Time Increasing Downward)**

Reading A File From Disk

In this simple example, let us first assume that you want to simply open a file (e.g., `/foo/bar`), read it, and then close it. For this simple example, let's assume the file is just 12KB in size (i.e., 3 blocks).

When you issue an `open("/foo/bar", O_RDONLY)` call, the file system first needs to find the inode for the file `bar`, to obtain some basic information about the file (permissions information, file size, etc.). To do so, the file system must be able to find the inode, but all it has right now is the full pathname. The file system must **traverse** the pathname and thus locate the desired inode.

All traversals begin at the root of the file system, in the **root directory** which is simply called `/`. Thus, the first thing the FS will read from disk is the inode of the root directory. But where is this inode? To find an inode, we must know its i-number. Usually, we find the i-number of a file or directory in its parent directory; the root has no parent (by definition). Thus, the root inode number must be "well known"; the FS must know what it is when the file system is mounted. In most UNIX file systems, the root inode number is 2. Thus, to begin the process, the FS reads in the block that contains inode number 2 (the first inode block).

Once the inode is read in, the FS can look inside of it to find pointers to data blocks, which contain the contents of the root directory. The FS will thus use these on-disk pointers to read through the directory, in this case looking for an entry for `foo`. By reading in one or more directory data blocks, it will find the entry for foo; once found, the FS will also have found the inode number of `foo` (say it is 44) which it will need next.

The next step is to recursively traverse the pathname until the desired inode is found. In this example, the FS reads the block containing the

ASIDE: READS DON'T ACCESS ALLOCATION STRUCTURES

We've seen many students get confused by allocation structures such as bitmaps. In particular, many often think that when you are simply reading a file, and not allocating any new blocks, that the bitmap will still be consulted. This is not true! Allocation structures, such as bitmaps, are only accessed when allocation is needed. The inodes, directories, and indirect blocks have all the information they need to complete a read request; there is no need to make sure a block is allocated when the inode already points to it.

inode of `foo` and then its directory data, finally finding the inode number of `bar`. The final step of `open()` is to read `bar`'s inode into memory; the FS then does a final permissions check, allocates a file descriptor for this process in the per-process open-file table, and returns it to the user.

Once open, the program can then issue a `read()` system call to read from the file. The first read (at offset 0 unless `lseek()` has been called) will thus read in the first block of the file, consulting the inode to find the location of such a block; it may also update the inode with a new last-accessed time. The read will further update the in-memory open file table for this file descriptor, updating the file offset such that the next read will read the second file block, etc.

At some point, the file will be closed. There is much less work to be done here; clearly, the file descriptor should be deallocated, but for now, that is all the FS really needs to do. No disk I/Os take place.

A depiction of this entire process is found in Figure 40.3 (page 11); time increases downward in the figure. In the figure, the open causes numerous reads to take place in order to finally locate the inode of the file. Afterwards, reading each block requires the file system to first consult the inode, then read the block, and then update the inode's last-accessed-time field with a write. Spend some time and understand what is going on.

Also note that the amount of I/O generated by the open is proportional to the length of the pathname. For each additional directory in the path, we have to read its inode as well as its data. Making this worse would be the presence of large directories; here, we only have to read one block to get the contents of a directory, whereas with a large directory, we might have to read many data blocks to find the desired entry. Yes, life can get pretty bad when reading a file; as you're about to find out, writing out a file (and especially, creating a new one) is even worse.

Writing A File To Disk

Writing to a file is a similar process. First, the file must be opened (as above). Then, the application can issue `write()` calls to update the file with new contents. Finally, the file is closed.

Unlike reading, writing to the file may also **allocate** a block (unless the block is being overwritten, for example). When writing out a new file, each write not only has to write data to disk but has to first decide

	data bitmap	inode bitmap	root inode	foo inode	bar inode	root data	foo data	bar data [0]	bar data [1]	bar data [2]
			read							
						read				
				read						
							read			
create (/foo/bar)		**read**								
		write								
							write			
					read					
					write					
				write						
					read					
write()	**read**									
	write									
								write		
					write					
					read					
write()	**read**									
	write									
									write	
					write					
					read					
write()	**read**									
	write									
										write
					write					

Figure 40.4: **File Creation Timeline (Time Increasing Downward)**

which block to allocate to the file and thus update other structures of the disk accordingly (e.g., the data bitmap and inode). Thus, each write to a file logically generates five I/Os: one to read the data bitmap (which is then updated to mark the newly-allocated block as used), one to write the bitmap (to reflect its new state to disk), two more to read and then write the inode (which is updated with the new block's location), and finally one to write the actual block itself.

The amount of write traffic is even worse when one considers a simple and common operation such as file creation. To create a file, the file system must not only allocate an inode, but also allocate space within the directory containing the new file. The total amount of I/O traffic to do so is quite high: one read to the inode bitmap (to find a free inode), one write to the inode bitmap (to mark it allocated), one write to the new inode itself (to initialize it), one to the data of the directory (to link the high-level name of the file to its inode number), and one read and write to the directory inode to update it. If the directory needs to grow to accommodate the new entry, additional I/Os (i.e., to the data bitmap, and the new directory block) will be needed too. All that just to create a file!

Let's look at a specific example, where the file `/foo/bar` is created, and three blocks are written to it. Figure 40.4 (page 13) shows what happens during the `open()` (which creates the file) and during each of three 4KB writes.

In the figure, reads and writes to the disk are grouped under which system call caused them to occur, and the rough ordering they might take place in goes from top to bottom of the figure. You can see how much work it is to create the file: 10 I/Os in this case, to walk the pathname and then finally create the file. You can also see that each allocating write costs 5 I/Os: a pair to read and update the inode, another pair to read and update the data bitmap, and then finally the write of the data itself. How can a file system accomplish any of this with reasonable efficiency?

THE CRUX: HOW TO REDUCE FILE SYSTEM I/O COSTS

Even the simplest of operations like opening, reading, or writing a file incurs a huge number of I/O operations, scattered over the disk. What can a file system do to reduce the high costs of doing so many I/Os?

40.7 Caching and Buffering

As the examples above show, reading and writing files can be expensive, incurring many I/Os to the (slow) disk. To remedy what would clearly be a huge performance problem, most file systems aggressively use system memory (DRAM) to cache important blocks.

Imagine the open example above: without caching, every file open would require at least two reads for every level in the directory hierarchy (one to read the inode of the directory in question, and at least one to read its data). With a long pathname (e.g., `/1/2/3/ ... /100/file.txt`), the file system would literally perform hundreds of reads just to open the file!

Early file systems thus introduced a **fixed-size cache** to hold popular blocks. As in our discussion of virtual memory, strategies such as **LRU** and different variants would decide which blocks to keep in cache. This fixed-size cache would usually be allocated at boot time to be roughly 10% of total memory.

This **static partitioning** of memory, however, can be wasteful; what if the file system doesn't need 10% of memory at a given point in time? With the fixed-size approach described above, unused pages in the file cache cannot be re-purposed for some other use, and thus go to waste.

Modern systems, in contrast, employ a **dynamic partitioning** approach. Specifically, many modern operating systems integrate virtual memory pages and file system pages into a **unified page cache** [S00]. In this way, memory can be allocated more flexibly across virtual memory and file system, depending on which needs more memory at a given time.

Now imagine the file open example with caching. The first open may generate a lot of I/O traffic to read in directory inode and data, but sub-

TIP: UNDERSTAND STATIC VS. DYNAMIC PARTITIONING

When dividing a resource among different clients/users, you can use either **static partitioning** or **dynamic partitioning**. The static approach simply divides the resource into fixed proportions once; for example, if there are two possible users of memory, you can give some fixed fraction of memory to one user, and the rest to the other. The dynamic approach is more flexible, giving out differing amounts of the resource over time; for example, one user may get a higher percentage of disk bandwidth for a period of time, but then later, the system may switch and decide to give a different user a larger fraction of available disk bandwidth.

Each approach has its advantages. Static partitioning ensures each user receives some share of the resource, usually delivers more predictable performance, and is often easier to implement. Dynamic partitioning can achieve better utilization (by letting resource-hungry users consume otherwise idle resources), but can be more complex to implement, and can lead to worse performance for users whose idle resources get consumed by others and then take a long time to reclaim when needed. As is often the case, there is no best method; rather, you should think about the problem at hand and decide which approach is most suitable. Indeed, shouldn't you always be doing that?

sequent file opens of that same file (or files in the same directory) will mostly hit in the cache and thus no I/O is needed.

Let us also consider the effect of caching on writes. Whereas read I/O can be avoided altogether with a sufficiently large cache, write traffic has to go to disk in order to become persistent. Thus, a cache does not serve as the same kind of filter on write traffic that it does for reads. That said, **write buffering** (as it is sometimes called) certainly has a number of performance benefits. First, by delaying writes, the file system can **batch** some updates into a smaller set of I/Os; for example, if an inode bitmap is updated when one file is created and then updated moments later as another file is created, the file system saves an I/O by delaying the write after the first update. Second, by buffering a number of writes in memory, the system can then **schedule** the subsequent I/Os and thus increase performance. Finally, some writes are avoided altogether by delaying them; for example, if an application creates a file and then deletes it, delaying the writes to reflect the file creation to disk **avoids** them entirely. In this case, laziness (in writing blocks to disk) is a virtue.

For the reasons above, most modern file systems buffer writes in memory for anywhere between five and thirty seconds, representing yet another trade-off: if the system crashes before the updates have been propagated to disk, the updates are lost; however, by keeping writes in memory longer, performance can be improved by batching, scheduling, and even avoiding writes.

TIP: UNDERSTAND THE DURABILITY/PERFORMANCE TRADE-OFF
Storage systems often present a durability/performance trade-off to users. If the user wishes data that is written to be immediately durable, the system must go through the full effort of committing the newly-written data to disk, and thus the write is slow (but safe). However, if the user can tolerate the loss of a little data, the system can buffer writes in memory for some time and write them later to the disk (in the background). Doing so makes writes appear to complete quickly, thus improving perceived performance; however, if a crash occurs, writes not yet committed to disk will be lost, and hence the trade-off. To understand how to make this trade-off properly, it is best to understand what the application using the storage system requires; for example, while it may be tolerable to lose the last few images downloaded by your web browser, losing part of a database transaction that is adding money to your bank account may be less tolerable. Unless you're rich, of course; in that case, why do you care so much about hoarding every last penny?

Some applications (such as databases) don't enjoy this trade-off. Thus, to avoid unexpected data loss due to write buffering, they simply force writes to disk, by calling `fsync()`, by using **direct I/O** interfaces that work around the cache, or by using the **raw disk** interface and avoiding the file system altogether[2]. While most applications live with the trade-offs made by the file system, there are enough controls in place to get the system to do what you want it to, should the default not be satisfying.

40.8 Summary

We have seen the basic machinery required in building a file system. There needs to be some information about each file (metadata), usually stored in a structure called an inode. Directories are just a specific type of file that store name→inode-number mappings. And other structures are needed too; for example, file systems often use a structure such as a bitmap to track which inodes or data blocks are free or allocated.

The terrific aspect of file system design is its freedom; the file systems we explore in the coming chapters each take advantage of this freedom to optimize some aspect of the file system. There are also clearly many policy decisions we have left unexplored. For example, when a new file is created, where should it be placed on disk? This policy and others will also be the subject of future chapters. Or will they?[3]

[2]Take a database class to learn more about old-school databases and their former insistence on avoiding the OS and controlling everything themselves. But watch out! Those database types are always trying to bad mouth the OS. Shame on you, database people. Shame.

[3]Cue mysterious music that gets you even more intrigued about the topic of file systems.

References

[A+07] "A Five-Year Study of File-System Metadata" by Nitin Agrawal, William J. Bolosky, John R. Douceur, Jacob R. Lorch. FAST '07, San Jose, California, February 2007. *An excellent recent analysis of how file systems are actually used. Use the bibliography within to follow the trail of file-system analysis papers back to the early 1980s.*

[B07] "ZFS: The Last Word in File Systems" by Jeff Bonwick and Bill Moore. Available from: `http://www.ostep.org/Citations/zfs_last.pdf`. *One of the most recent important file systems, full of features and awesomeness. We should have a chapter on it, and perhaps soon will.*

[B02] "The FAT File System" by Andries Brouwer. September, 2002. Available online at: `http://www.win.tue.nl/%7eaeb/linux/fs/fat/fat.html`. *A nice clean description of FAT. The file system kind, not the bacon kind. Though you have to admit, bacon fat probably tastes better.*

[C94] "Inside the Windows NT File System" by Helen Custer. Microsoft Press, 1994. *A short book about NTFS; there are probably ones with more technical details elsewhere.*

[H+88] "Scale and Performance in a Distributed File System" by John H. Howard, Michael L. Kazar, Sherri G. Menees, David A. Nichols, M. Satyanarayanan, Robert N. Sidebotham, Michael J. West.. ACM TOCS, Volume 6:1, February 1988. *A classic distributed file system; we'll be learning more about it later, don't worry.*

[P09] "The Second Extended File System: Internal Layout" by Dave Poirier. 2009. Available: `http://www.nongnu.org/ext2-doc/ext2.html`. *Some details on ext2, a very simple Linux file system based on FFS, the Berkeley Fast File System. We'll be reading about it in the next chapter.*

[RT74] "The UNIX Time-Sharing System" by M. Ritchie, K. Thompson. CACM Volume 17:7, 1974. *The original paper about UNIX. Read it to see the underpinnings of much of modern operating systems.*

[S00] "UBC: An Efficient Unified I/O and Memory Caching Subsystem for NetBSD" by Chuck Silvers. FREENIX, 2000. *A nice paper about NetBSD's integration of file-system buffer caching and the virtual-memory page cache. Many other systems do the same type of thing.*

[S+96] "Scalability in the XFS File System" by Adan Sweeney, Doug Doucette, Wei Hu, Curtis Anderson, Mike Nishimoto, Geoff Peck. USENIX '96, January 1996, San Diego, California. *The first attempt to make scalability of operations, including things like having millions of files in a directory, a central focus. A great example of pushing an idea to the extreme. The key idea behind this file system: everything is a tree. We should have a chapter on this file system too.*

Homework (Simulation)

Use this tool, `vsfs.py`, to study how file system state changes as various operations take place. The file system begins in an empty state, with just a root directory. As the simulation takes place, various operations are performed, thus slowly changing the on-disk state of the file system. See the README for details.

Questions

1. Run the simulator with some different random seeds (say 17, 18, 19, 20), and see if you can figure out which operations must have taken place between each state change.
2. Now do the same, using different random seeds (say 21, 22, 23, 24), except run with the `-r` flag, thus making you guess the state change while being shown the operation. What can you conclude about the inode and data-block allocation algorithms, in terms of which blocks they prefer to allocate?
3. Now reduce the number of data blocks in the file system, to very low numbers (say two), and run the simulator for a hundred or so requests. What types of files end up in the file system in this highly-constrained layout? What types of operations would fail?
4. Now do the same, but with inodes. With very few inodes, what types of operations can succeed? Which will usually fail? What is the final state of the file system likely to be?